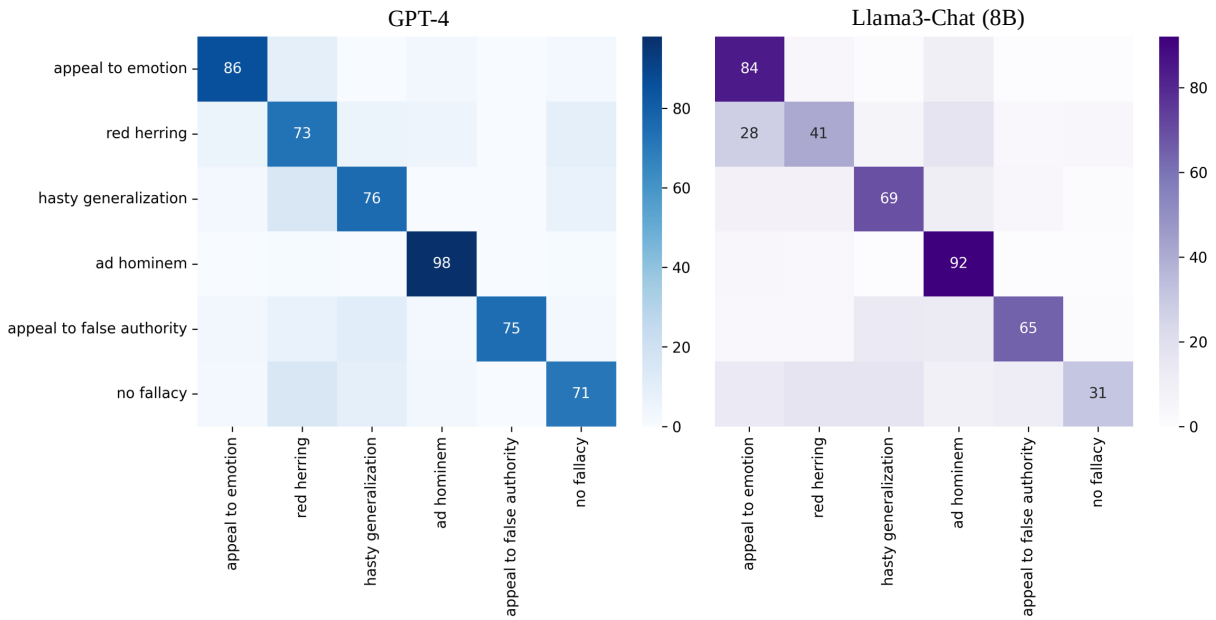
D.2 Error Cases

Scheme	Model		
DG	GPT-4	**Input**	A: Do we need a global environment task force? B: The global taskforce for human rights is doing a good job. That's why we need one for the environment, too.
		Output	Hasty Generalization: A conclusion based on insufficient or biased evidence; rushing to a conclusion before you have all relevant facts.
		Pred.	No Fallacy
		GT	Hasty Generalization
	Mistral	**Input**	<YouTube conspiracy theorist> Jake Morphonios said that for all he knows, Doug Haig didn't play a role in Paddock's massacre.
		Output	Red Herring: Introducing a distracting or irrelevant topic, so as to lead others away from the main subject at hand, and avoid addressing the actual argument. Name-calling Using derogatory or offensive terms to describe someone or their arguments, instead of presenting a counterargument.
		Pred.	Red Herring
		GT	Name-calling
GFA	GPT-3.5	**Input**	something that kills is deadly.
		Output	The focal argument of the discourse is logically reasonable. The statement "something that kills is deadly" is a tautology, as it simply states that if something has the capability to kill, it is indeed deadly. This argument is logically sound and does not contain any fallacies.
		Pred.	No Fallacy
		GT	Circular Reasoning
	Llama-2	**Input**	If you are outraged by this, stop eating meat and dairy. Otherwise, you are a hypocrite
		Output	The comment's argument is potentially fallacious. The argument is an example of the "hasty generalization" fallacy, where a conclusion is drawn based on insufficient or incomplete evidence. The comment assumes that anyone who is outraged by the topic must be a hypocrite if they consume meat and dairy, without considering the possibility that they may have valid reasons for their outrage that have nothing to do with their dietary choices. Additionally, the comment does not address the possibility that some people may be vegetarian or vegan and still be outraged by the topic. Therefore, the comment's argument is not logically reasonable and may be considered fallacious.
		Pred.	Hasty Generalization
		GT	False Dilemma
GFA-W	GPT-3.5	**Input**	"<Are the 2,000 U.S. troops in Syria really up to that task,> should the Turks decide to cleanse the Syrian border of Kurds, or should the Syrian regime decide to take back territory occupied by the Kurds?"
		Output	The fragment highlighted by '<>' is potentially fallacious, specifically exhibiting a form of hasty generalization. It questions whether 2,000 U.S. troops are sufficient to handle major conflicts without providing detailed evidence or context about the capabilities of these troops, their operational support, or the nature of the potential military engagements they might face. This can lead to an oversimplified conclusion about the effectiveness of U.S. military presence based on troop numbers alone, without considering other strategic factors.
		Pred.	Causal Oversimplification
		GT	Doubt Credibility
	Mistral	**Input**	Paige wants to play a guitar solo on our next recording. Her father is a horrible musician, though, so I doubt that Paula's any good either.
		Output	The argument presented is potentially fallacious, known as "Negative Ad Hominem Fallacy", which is an argument that attacks the person rather than the argument, and incorrectly assumes that because the father is a "horrible musician", his daughter (Paige) must be equally "horrible" as well. It is not fair to judge the skills of a person (Paige) based on the skills of another person (her father) without considering Paige's own talents or skills.
		Pred.	Red Herring
		GT	Hasty Generalization
CoT	GPT-4	**Input**	"I have contacted the school's principal, Dr.", "Branch, and I trust that he will take this matter seriously and deal with it appropriately." "Is the principal acting on this?"
		Output	The fallacy present in the segment highlighted by '<>' is "Red Herring." This is because the question about the principal's action diverts attention from the main issue discussed in the article (Islamizing the schools) to the actions of the school principal, which is a separate issue.
		Pred.	Red Herring
		GT	Doubt Credibility
	Llama-3	**Input**	I don't think you can just rely on U.N. sanctions, Randee. But you're absolutely correct, it is a threat, it's a huge threat.
		Output	After analyzing the speech, I found that the fallacy present in the highlighted argument is the False Causality (Post Hoc Fallacy). The speaker is implying that the threat from North Korea grew while the President was preoccupied with Iraq, suggesting that the President's focus on Iraq caused the threat to grow. However, this is a post hoc fallacy because it's an assumption that the President's focus on Iraq caused the threat to grow, without providing any evidence or logical connection between the two.
		Pred.	False Causality

		GT	Appeal to Emotion
P&C	GPT-4	**Input**	I bought a ticket to win a new car at the mall, since I have never won anything like that in the past."",
		Output	The premise that the person has never won anything like a new car in the past does not logically entail the conclusion that they bought a ticket to win a new car at the mall. The premise provides a context or a possible motivation for why they might have decided to buy a ticket, perhaps hoping to change their luck, but it does not necessarily lead to the conclusion that buying a ticket was a rational or necessary action. There is no causal or logical necessity connecting the lack of previous winnings to the decision to buy a ticket; the conclusion does not directly follow from the premise.
		Pred.	No Fallacy
		GT	False Causality
	Llama-3	**Input**	A: Is Turkey ready to join the European Union? B: Turkey helped us with the Refugees. They were beneficial for the EU there, they will be beneficial for the EU in general.
		Output	The premise "Putin has nothing to say regarding the EU. It is our choice who joins and who does not" does not entail the conclusion "Turkey is ready to join the European Union". This premise only establishes that the decision to admit Turkey into the EU is up to the EU member states, not Putin, and does not provide any information about Turkey's readiness to join. The premise "Turkey and Europe share much culture and many people from Turkey already live in European countries. They are ready" is more relevant to the conclusion, but it is still a weak argument. The fact that Turkey and Europe share culture and have people living in each other's countries does not necessarily mean that Turkey is ready to join the EU. There are many other factors to consider, such as Turkey's political system, economic stability, and human rights record, among others. Therefore, the premises do not entail the conclusion, and B's argument is flawed.
		Pred.	Appeal to Consequences
		GT	Hasty Generalization

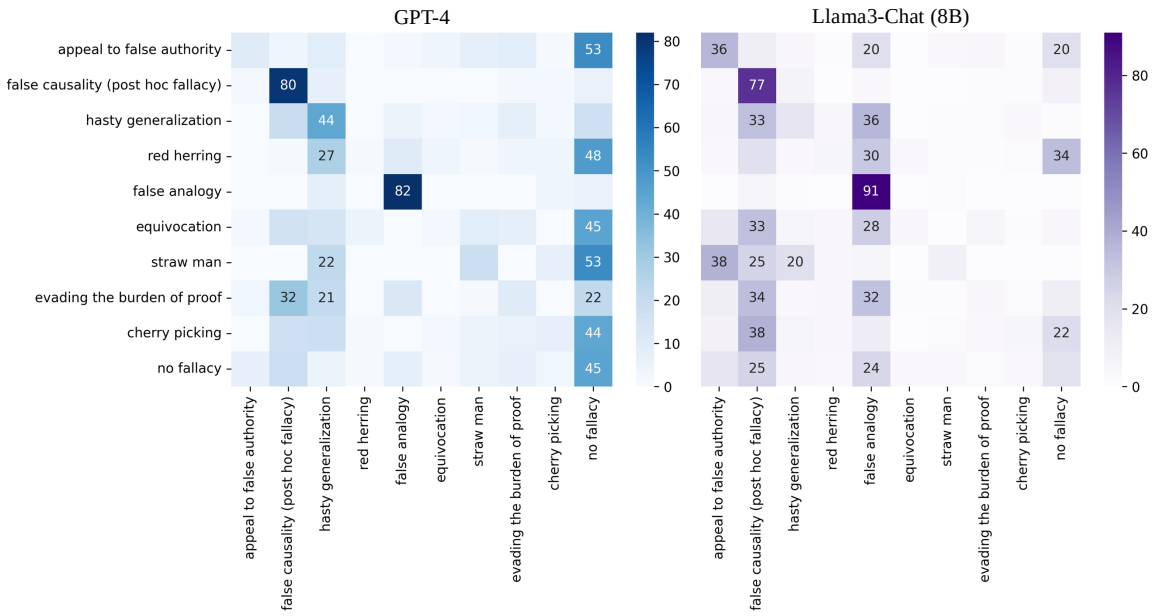
Table 30: Error cases of GPT-4 and Llama3-Chat (8B) for each multi-round prompting scheme.

D.3 Misclassification Confusion Matrices of GPT-4 and Llama3

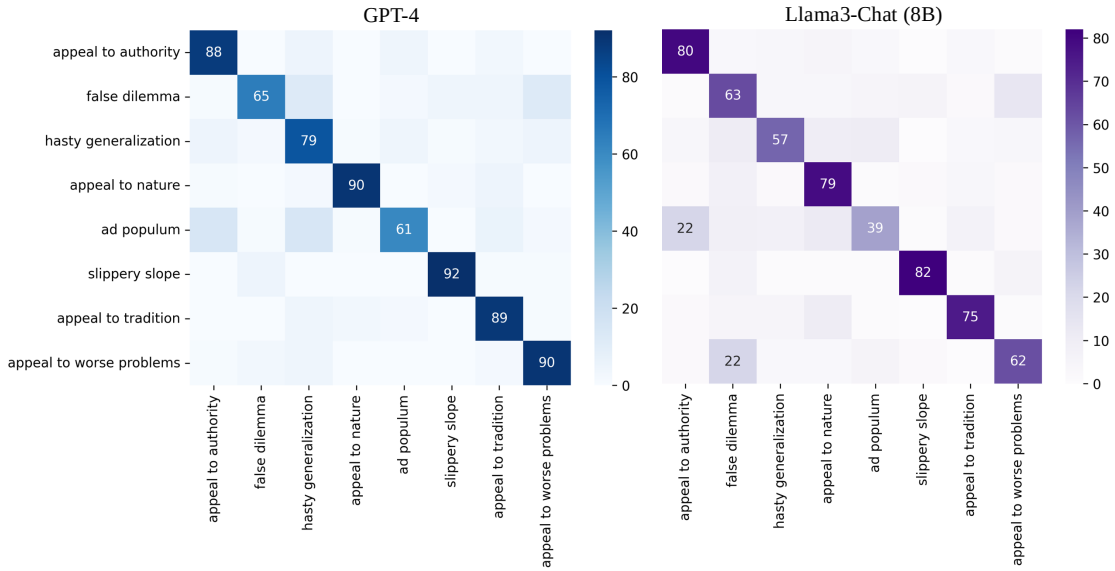


(a) Argotario

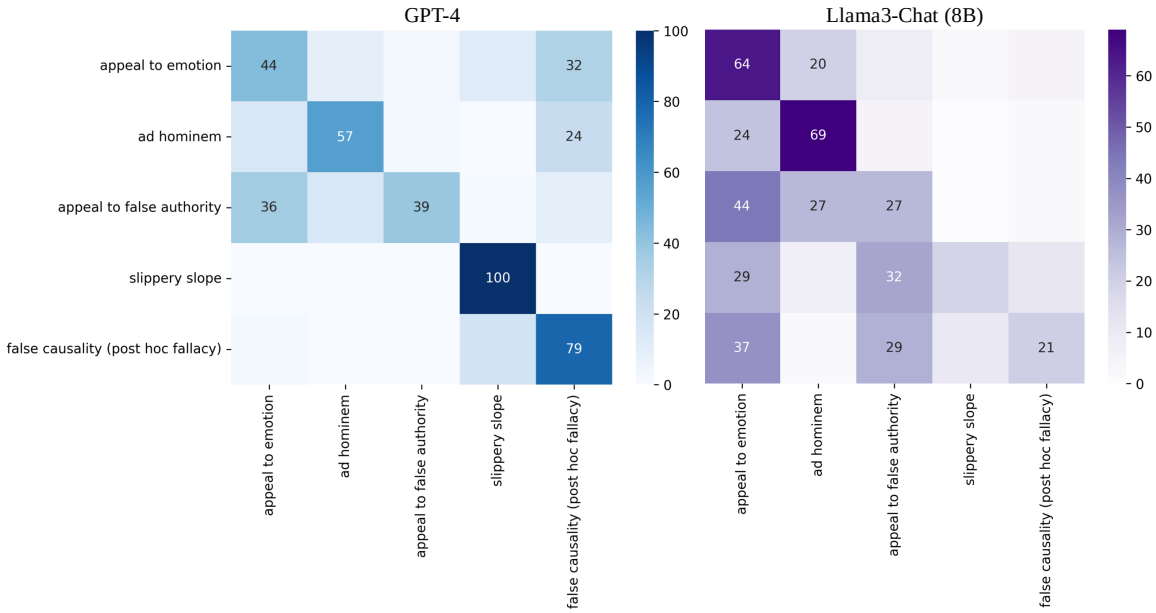
Figure 7: Misclassification confusion matrix of GPT-4 and Llama3-Chat (8B) on ARGOTARIO. Cell values are the percentages of row fallacy types that are misclassified as column fallacy types.



(b) Covid-19



(c) Reddit



(d) ElecDeb

Figure 8: Misclassification confusion matrix of GPT-4 and Llama3-Chat (8B) on COVID-19, REDDIT and ELECDEB. Cell values are the percentages of row fallacy types that are misclassified as column fallacy types.